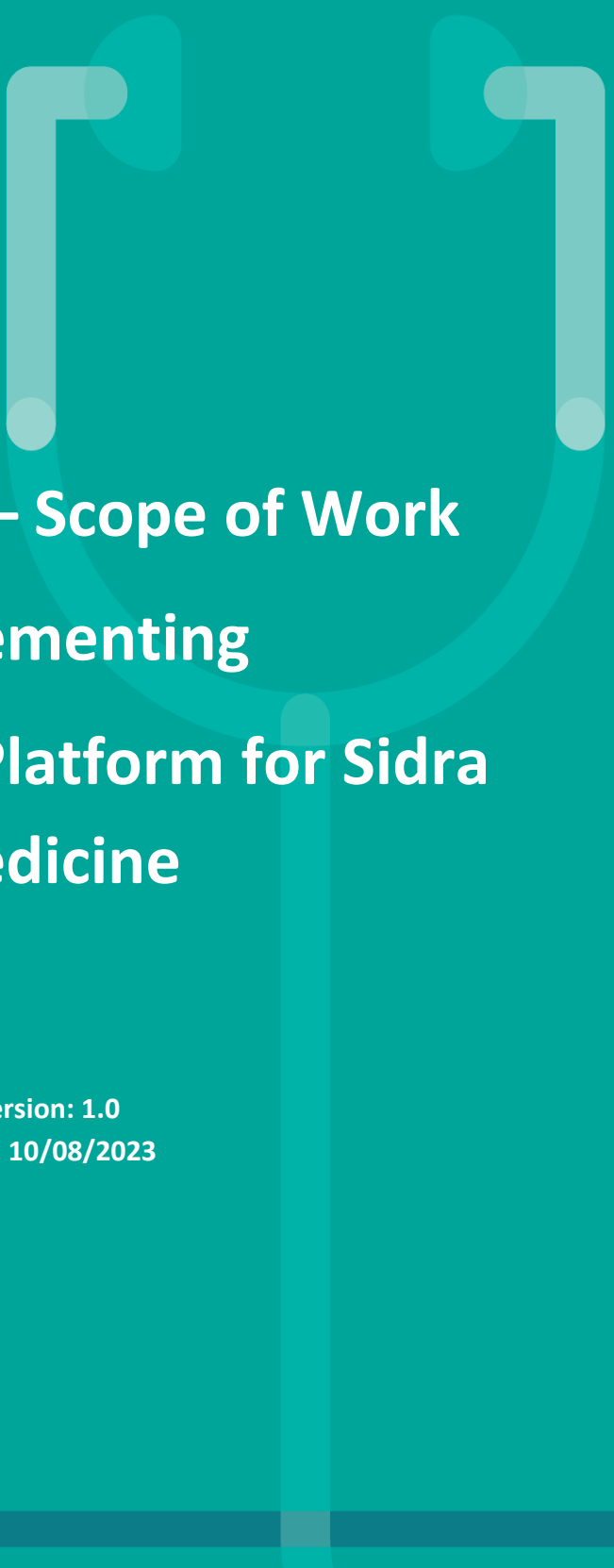




Appendix A – Scope of Work

Implementing

A Data-Lake Platform for Sidra

Medicine

Version: 1.0
Date: 10/08/2023

Revision History

VERSION NO.	REVISION DATE	CHANGE REFERENCES & SUMMARY
1.001	10/08/2023	Initial Draft 1.0
1.1	01/7/2024	

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1. Introduction

Sidra Medicine vision is to become a beacon of learning, discovery, and exceptional care, ranked among the top academic medical centers in the world. One of Sidra's design principles is being a fully digital hospital that utilizes information, data, and insights to drive patient-care, clinical outcomes, research and operational efficiency.

Currently, there are several applications and platforms deployed at Sidra Medicine such as Cerner Electronic Medical Records (Cerner EMR), Oracle Fusion Cloud ERP, Datix, Qualtrics and others which support daily clinical and administrative operations of the hospital and digitize patients' journey. Those applications generate massive volumes of data encompassing patient records, administrative information, financial transactions, quality related incidents, patient surveys and others. In addition, there are multiple Research platforms that generate data related to Whole Genome Sequencing (WGS) and Genome reference data, Biorepository, phenotype data, clinical trials and clinical registries.

The dispersed and siloed nature of this data often hampers Sidra Medicine ability to timely extract meaningful insights, hindering swift decision-making and optimization of resources as significant number of man-hours spent on manually extracting reports from different applications as well as manually distribute them to different users.

By building a Data Lake and analytics solutions, Sidra Medicine aims to create a single source of truth and uncover actionable insights, predict trends, and align their strategies with real-time operational metrics. This empowers Sidra's Clinicians, Researchers and Executives to make informed, data-driven decisions that foster improved patient care, elevate financial and operational performance, and position the hospital at the forefront of healthcare innovation.

2. Objectives

The primary objective of this project is to build a robust Data Lake and Analytics platform that enables efficient data extraction, transformation, and analysis of different data sources across Sidra Medicine.

Below are the business objectives and benefits that would be realized by the end of the project implementation:

Single Source of Truth: Consolidate disparate data sources into a single data lake.

Improved Decision-Making: Access to timely, accurate, and comprehensive insights to enable data driven decision-making at all levels of the organization, leading to better outcomes and strategies.

Operational Efficiency: Automated data extraction, transformation, and loading (ETL) processes reduce manual data entry and manipulation efforts, saving time and minimizing errors.

Strategic Planning and Budgeting: Accurate and comprehensive data supports effective strategic planning, budgeting, monitoring and allocation of resources for future growth and initiatives.

Effective Patient Care: Hospital staff can access and utilize a unified dashboard related to patient care, cost of treatment, patient experience and staffing utilization which aiding in care optimization and efficiency.

Unified Reporting and Analysis: Sidra executives, clinical and research staff can access integrated reports and dashboards that provide a holistic view of operations, finances, patient care, and other critical aspects.

Enhanced Performance Measurement: Key performance indicators (KPIs) can be monitored consistently, allowing the hospital to assess its performance and align activities with strategic objectives.

Reduced Data Latency: Real-time or near-real-time data updates allow for faster reporting, making critical information available for time-sensitive decisions.

Self-Service Analytics: Hospital staff can use user-friendly tools to perform ad hoc queries and analyses without needing extensive technical skills, promoting self-service data exploration and analysis.

Data Visualization: Interactive visualizations and dashboards provide intuitive ways to communicate complex data, enabling quicker understanding of trends and insights.

Data Governance and Security: The Data-Lake allows for centralized data governance, ensuring data security, privacy, access control, and compliance with regulatory standards and best practices (i.e. Role Based access control and data encryption).

Data Catalog and Metadata Management: The data-lake can provide a centralized repository to catalog and manage the metadata associated with the stored data.

Long-Term Data Retention: The data-lake provides a secure and accessible repository for historical data, ensuring valuable insights are retained even as ERP systems evolve.

Predictive Analytics, AI, and Machine Learning: The data-lake solution can provide a rich dataset and platforms for training predictive models, AI, and machine learning algorithms, and support data science workloads. It includes tools and frameworks for statistical analysis, machine learning, and AI.

Precision Medicine and Personalized Healthcare: The data-lake can support precision medicine initiatives by integrating genomics data, molecular profiling, and clinical data to identify biomarkers, develop targeted therapies, and personalize treatment plans based on an individual's unique characteristics and genetic makeup.

3. Scope of Work and Deliverables

The scope of work includes the following:

1. Requirements gathering and analysis
2. Data-Lake / Data-Lakehouse architecture design
3. Infrastructure setup
4. Data ingestion and integration (both batch and real-time).
5. Data processing and transformation
6. Data storage and management
7. Data Governance and Compliance
8. Analytics, Reporting and Machine Learning models setup
9. Testing and validation
10. Documentation and training

The project major tasks and deliverables include the following:

3.1 Project Initiation and Requirements Gathering

- Identify stakeholders and establish communication channels.
- Identify the list of data sources.
- Conduct meetings with stakeholders to gather data requirements, business objectives, analytics needs and Machine Learning use cases.
- Document stakeholder inputs and finalize the scope of the project including functional and non-functional requirements.
- Deliverable: Requirements Specification Document.

3.2 Data Lake / Lakehouse Architecture Design

- Design a robust data lake architecture that includes:
 - Raw Data Zone: Ingest raw data from various sources.
 - Cleansed Data Zone: Store cleansed and transformed data.
 - Curated Data Zone: Organize data for specific analytical use cases
- Design a specific subject areas Data Marts for detailed analysis.
- **Tools and Technologies**
Utilize the following Azure services:
 - Azure Databricks: For data processing and analytics.

- Azure Data Lake Storage (ADLS): For scalable data storage.
- Azure Synapse Analytics: For data warehousing.
- Azure Data Factory: For data integration and orchestration.
- Azure Purview: for unified data governance
- Deliverable: detailed Data Architecture Designing document

3.3 Infrastructure Setup on Azure

▪ Resource Provisioning

Provision and configure the following resources:

- Azure Databricks workspace
- Azure Data Lake Storage
- Azure Synapse Analytics

▪ Network Configuration

Ensure secure and efficient network setup:

- Virtual Network setup
- Network Security Groups
- Private Endpoints for secure data access
- Deliverable: Complete configuration and environment setup sign-off according to architectural design and best practice.

3.4 Data Ingestion and Integration

▪ Data Ingestion

Implement data ingestion pipelines:

- Batch data ingestion using Azure Data Factory
- Real-time data ingestion using Azure Event Hubs and Stream Analytics

▪ Data Integration, Processing and Transformation

- Integrate data through ETL/ELT processes and perform data cleansing and transformation.
- Develop an efficient and optimized ETL/ELT pipeline.

Deliverable: complete data integration and ETL/ELT pipelines.

3.5 Data Quality and Validation

- Implement data quality checks and validation rules to ensure accuracy and consistency.
- Develop processes to identify and resolve data anomalies, duplicates, and discrepancies.
- Establish data profiling and monitoring mechanisms to ensure ongoing data quality.
- Deliverable: Data Quality and Validation Framework Implementation

3.6 Data Storage and Management

- Organize data in Azure Data Lake Storage

- Optimize data storage in Azure Synapse Analytics (Table design, indexing and partitioning)

3.7 Data Governance and Compliance

- Implement robust security measures and role-based access control.
- Implement core master data management capabilities.
- Implement Data Catalog and metadata management.
- Ensure the implementation compliance with local and international data security and privacy laws and frameworks.

3.8 Analytics, Reporting and Machine Learning Models Platforms

- Ensure seamless integration with Sidra's existing PowerBI platform.
- Develop dashboards and operational reports for Finance, HR, Supply Chain and other subject areas covered by Oracle Fusion ERP data source.
- Configure Databricks SQL Analytics for querying
- Re-map existing PowerBI dashboards to the newly developed Data Marts.
- Configure Machine Learning platform and build few data models for both Clinical and Non-Clinical subject areas.

3.9 Testing and Quality Assurance

- Perform unit testing of ETL scripts, data lake and data-warehouse components.
- Conduct integration testing to ensure seamless integration and end to end functionality.
- Validate Data-Lake accuracy and completeness against source systems data.
- Engage stakeholders in user acceptance testing (UAT) to ensure the solution meets business requirements.
- Deliverable: Test Plans and Test Results

3.10 User Training and Documentation

- Prepare user training materials and guides for data engineers and administrators.
- Develop standard operating procedures documents
- Prepare admin and support guides for the entire Data Lake Platform.
- Conduct training sessions for developers, engineers and end users.
- Create documentation for data dictionaries, ETL processes, data lineage, and best practices.
- Deliverable: User Training Materials and Documentation

3.11 Deployment and Go-Live

- Plan the deployment of the Data Lake Platform to the production environment.
- Migrate the ETL processes, data, and configurations to the production environment.
- Monitor system performance during the initial period of usage.

- Address any issues or concerns that arise during deployment.
- Deliverable: Production Data Warehouse and Analytics Environment

3.12 Post-Implementation Support

- Provide ongoing technical support to address any system issues, bugs, or enhancements.
- Monitor system performance, data quality, and user interactions.
- Implement updates and modifications based on evolving business requirements.
- Deliverable: Support and Maintenance Documentation

4. Software Licenses

The vendor should be able to develop and implement the solution using a software stack currently available at Sidra Medicine without additional licensing cost. Please refer to the below technology stack available:

- Azure Databricks
- Azure Data Factory
- Azure Synapse Analytics
- Microsoft PowerBI
- Azure Machine Learning or Databricks Mosaic ML
- Other Azure platforms available within Azure Qatar data center.

If additional software licenses are required, it needs to be mentioned clearly with the projected licensing cost.

5. Timeline and Milestones:

The project is estimated to be completed within 10 months following the major activities and deliverables listed in section 3 above.

The vendor should provide a detailed project plan indicating the project management methodology they will follow.

Activity/Deliverable	Delivery Schedule (from start of engagement)
Requirements gathering and analysis	Month 1
Data-Lake / Data-Lakehouse architecture design	Month 1
Infrastructure setup	Month 1 – Month 2
Data ingestion and integration (both batch and real-time).	Month 1 – Month 5
Data processing and transformation	Month 2 – Month 7

Data storage and management	Month 2– Month 8
Data Governance and Compliance	Month 3 – Month 9
Analytics, Reporting and Machine Learning models setup	Month 7 – Month 9
Testing and validation	Month 7 – Month 9
Documentation and training	Month 8 – Month 10

6. Vendor Supplied Resources

- Unless provided otherwise elsewhere in the SOW, Vendor shall provide all the resources (manpower, equipment, tools, etc.,) required to execute and accomplish the project.
- Vendor shall provide manpower resources with the required skills set
- Vendor resources should include the following roles (at minimum):
- **Project Manager:** with experience in managing large scale Data Lake and DataWarehouse projects and project management qualifications (ie. Min exp 7+ years with certified project management professional PMP, Certified Scrum Master CSM, Agile PMP and others).
- **Data Engineers:** with experience in developing and delivering large scale Databricks implementation. Certificates from Databricks and Microsoft Azure Data Engineering will be highly desirable. (Min exp 7+ years in Data Engineering)
- **Data Architects:** Experts in data modelling and data architecture for large scale data lake implementations and act as subject matter experts. Related certifications in data modelling will be highly desirable. (Min exp 7+ years in Data Modelling)
- **ML Engineer/Data Scientist:** to implement, optimize and deploy machine learning models driven by different use cases identified on the requirement field. Experience in AI and data science implementation and related certification in Azure ML will be highly desirable.
- **SQL Developers & PowerBI developers:** to work closely with Sidra's team and remap existing PowerBI dashboards to the newly developed data marts. Also to develop new dashboards linked to Oracle Fusion ERP subject area. (Min exp 3+ years in PowerBI & SQL)
- **Oracle Fusion ERP Data Expert:** to act as a subject matter expert for Oracle Fusion ERP data source and guide the project technical team on best practices related to data extraction, modelling and reports development. To work closely with our business users (Finance, HR and SCM) and identify best dashboards and reports to support operational and strategic decision making. (Min exp 5+ years in Oracle Fusion)
- **Data Governance Expert:** to implement data governance related standards and technologies as identified in the scope. Experience in Data Quality, Master Data Management and Data Catalog is a must. Certificates in Data Governance is highly desirable. (Min exp 10+ years in Data Governance)
- Vendor shall provide CVs and profile for the individuals who will be working on delivering the project and covering the above roles and others.

7. Sidra Supplied Resources

- Office space complete with furniture (if the vendor's resources are required on-site)
- Access to internet (if the vendor's resources are required on-site)
- Access to existing data servers and dashboards

8. Duration of the Agreement

Vendor shall complete the SERVICES within 10 months from the project kick off date.,

9. Status Meetings and Reports

the Vendor's Agreement Manager, the Sidra Senior or other Program Manager and other appropriate representatives of the Parties shall meet at a mutually agreed site or by teleconference, as requested by Sidra, to discuss the status of the project. During such status meetings, the Parties shall discuss, the status of the project and any difficulties or issues that may exist, and any proposed changes to dates generally and any Due Dates specifically. Vendor shall keep minutes of the meetings and shall issue minutes in draft form to all meeting attendees promptly following each meeting. Following its receipt thereof, the recipient promptly shall review and provide to the drafter any feedback and comments it may have with respect to such minutes, following which the drafter shall incorporate any feedback and comments into the minutes and re-issue the minutes in final form. At least two (2) business days before each such status meeting, Vendor shall present to Sidra a written report of the status of the project. The report shall include a summary in such detail as Sidra shall reasonably request of the accomplishments and difficulties encountered, suggestions and proposed actions for dealing with and resolving any identified difficulties and the anticipated results, and a comprehensive and consolidated log of all outstanding project-related problems identified by Sidra and Vendor that remain to be resolved. Sidra shall have the right to assume that Vendor is not aware of any problems, issues or risks unless Vendor specifically identifies such problems, issues or risks in the project status reports.

10. Pre-Live testing

Vendor shall arrange Pre-live testing ("**Pre-Live Testing**") in accordance with applicable Test Plan to be agreed by the parties. The Test Plan shall specify the relevant Acceptance Criteria including Data Quality and Dashboards performance metrics, and the sequence and the timings to be followed. Sidra shall report to Vendor any defects discovered during Pre-Live Testing, and Vendor promptly shall correct such defects. When Pre-Live Testing has been successfully concluded, Sidra shall so notify Vendor , and the Software thereafter will be ready for Production Use.

11. Production use

Following successful completion of Pre-Live Testing, Sidra will deploy the Software in production. During the Production Use Period, Vendor shall work with Sidra to stabilize, and then optimize, the use of the Software. If any defects are discovered during the

Production Use Period, Sidra shall report such defects to Vendor, and Vendor promptly shall correct such defects. When the Production Use Period has been successfully concluded (all defects have been corrected), Sidra shall so notify Vendor.

12. Project Completion and Hand-Over

Upon successful completion of the project, the project manager will facilitate a thorough review and verification of all deliverables to ensure they meet the defined acceptance criteria.

Each of the following conditions are required for a project to achieve "**Project Completion**": (a) the Production Use Period has been successfully completed; (b) Vendor has provided to Sidra all Documentation and other deliverables relating to the project. Nothing else, including Sidra's use of the Software, or any component thereof, in a live, production environment shall constitute Project Completion, affect any rights and remedies that may be available to Sidra and/or constitute or result in "acceptance" under general Agreement law or any other Law.

If the WORK has been performed and completed in accordance with the AGREEMENT, Sidra will issue a CERTIFICATE OF COMPLETION stating the COMPLETION DATE therein. Thereafter, Vendor shall commence providing Maintenance and Support services.

13. Warranty period for the software

The Warranty Period for the WORKS performed shall be twelve (12) months from project COMPLETION DATE, and VENDOR shall warrant that the WORKS shall be free from defects, errors and bad workmanship throughout the warranty period.

14. Evaluation Criteria

The vendor must provide evidence of having the following:

Functional & Operational Expertise:

- Evidence of previous experience in large-scale Datawarehouse and Data-Lake implementation.
- Evidence of deep expertise in data architecture, data modelling and ETL/ELT development.
- Evidence of previous implementation of Analytics solutions and DataWarehouse that is integrated with Oracle Fusion ERP data source
- Proven experience in large-scale project management using Agile framework.
- Evidence of previous implementation in Machine Learning and Data Science domain.
- Experience working in healthcare domain or related industries is preferred.

Technical Expertise:

- Deep knowledge of Data-Lake technologies in general and specifically Azure Databricks and Azure Data Factory stack.
 - Proven expertise in developing and deploying large scale Analytics projects.
 - Deep knowledge and proven expertise in Data Architecture and ETL/ELT best practices, implementation, optimization and support.
 - Deep knowledge proven expertise in Data Governance standards and best practices.
 - Experience in visual analytics and dashboards development using PowerBI platform.
 - Vendor should be listed as Microsoft and Databricks official partner (Gold Partner preferred).
-
- Proposal should be very clear and self-explanatory with a straightforward, concise description for each component/deliverable as listed in this RFP request.
 - Delivery mode for the project shall be hybrid with on-site technical engineers and resources throughout the project implementation and post go-live support.
 - Proposal should provide all the technical details related to Data Architecture, ETL/ELT technologies and ML platforms.
 - The proposal should clearly define how the platform will be scalable to handle increased data volume and user loads in the future as well as accommodate new data sources and systems.
 - The proposal should define clear metrics that will be used for data processing, storage and retrieval times.
 - The vendor must have done similar implementations in other organizations
 - The vendor must provide proper references and sample reports for such implementations.
 - The vendor must provide all the documentations required as stated in the deliverable section.
 - The vendor must comply with all technical requirements and information security guidelines.

15. RESPONSE REQUIREMENTS

The proposal should include the following elements and should be submitted in a word document format/a PDF along with embedded diagrams (where applicable, process flows, functional system design schema):

#	Response Title	Description/Format						
1	Company profile	- Provide past and present projects including client name, project region and industry, scope of implementation and support services, contract duration, contract value, and project references. - Achievements and Awards: Highlights of the company's accomplishments, such as awards received, industry recognition, or notable projects completed. - Overview of the company's organizational hierarchy, including key executives, departments, and reporting lines.						
2	Commercial and Financial Information	Submit the last 3 years of audited financial statements and valid commercial registration status.						
3	Valid commercial registration							
4	Technical Proposal	Requirement delivery plan - Address all functional and technical requirements outlined in the SOW. - Proposed architectural solution that seamlessly integrates with Sidra's existing infrastructure.						
		Data Migration Plan - Submit a high-level plan outlining the migration process, including phases, testing procedures, security measures, data cleansing and transformation methods, and validation procedures. - This response should address the requirements outlined in this SOW.						
		Integration Plan - Document support for integrations, ensuring compliance with security features, local healthcare industry standards, standard communication protocols, and scalability to cloud implementations. - This response should address the requirements outlined in this SOW.						
		Security Management Plan Detail how the proposed solution complies with Sidra Medicine's security and privacy controls, including access control mechanisms, in addition to responses to the Cloud Security Questionnaire						
		Training and Support Plan Include a technical support plan, service level agreement (SLA), training and knowledge transfer documentation, customization capabilities, and user community support resources. This response should address the requirements outlined in this SOW.						
		Warranty Coverage Discuss the extent of warranty coverage provided, including any exclusions.						
		Project Plan with timelines (In gantt Chart/MS project), that includes the following activities <table><tr><th>Phase</th><th>Tasks</th></tr><tr><td>Requirements Definition and Analysis</td><td>- Stakeholder consultations and user interviews</td></tr><tr><td></td><td>- Document functional and non-functional requirements</td></tr></table>	Phase	Tasks	Requirements Definition and Analysis	- Stakeholder consultations and user interviews		- Document functional and non-functional requirements
		Phase	Tasks					
		Requirements Definition and Analysis	- Stakeholder consultations and user interviews					
	- Document functional and non-functional requirements							

		Design and Prototyping	- Develop system architecture, user interfaces, and data migration plan
			- Create prototypes and conduct proof of concept
		Development and Testing	- Develop core system functionalities on sandbox environment
			- Conduct unit testing and integration testing
		User Acceptance Testing (UAT)	- Prepare test scenarios and conduct UAT sessions
		Deployment and Training	- Prepare deployment plan and deploy system to production
			- Conduct user training sessions
		Documentation and Knowledge Transfer	- Update documentation and knowledge transfer materials
			- Conduct knowledge transfer sessions with stakeholders and support teams
		OTHERS - List of resources (and their cvs) that vendor will utilize/mobilize for the implementation of the project - CV of proposed project manager - Project organizational chart	